

BE FHIR-a-thon - Test atelier proposal

From forms to FHIR: definition-based extraction in Belgian healthcare

Objective

This atelier demonstrates and tests definition-based extraction - a mechanism in FHIR SDC (Structured Data Capture) that automatically transforms completed QuestionnaireResponses into discrete, interoperable FHIR resources without custom per-vendor mapping code.

By linking Questionnaire items to StructureDefinitions via the `.definition` element, the `$extract` operation converts form responses into Bundles of profiled FHIR resources - making the Questionnaire both form specification and extraction specification.

Scope & Use cases

Two Belgian use cases anchor the atelier:

- **Thuishospitalisatie:** Home care nurse documents an OPAT treatment or antitumoral therapy. Extracted resources (Observations, DiagnosticReport, MedicationStatements) flow into the patient record. Directly linked to the FOD-funded thuishospitalisatie pilot (UZ Leuven, Corilus, Wit-Gele Kruis, nexuzhealth, running since January 2026).
- **Registry population:** Clinician submits a QERMID implant registration or cancer notification (BCR). Output is validated against registry StructureDefinitions and forwarded to the receiving infrastructure. Relevant for Sciensano/HealthData.be (HD4DP), Belgian Cancer Registry, and BSP.

Three Roles in the Data Flow

Role	Actor	Responsibility	Our partner in this testatelier
Domain expert	Registries & institutions	Publish StructureDefinitions (profiles / logical models); co-author Questionnaire .definition links.	UZ Leuven
Data provider	EHR vendors, care software	Render Questionnaire, capture QuestionnaireResponse, call \$extract (shared API available - no own implementation required).	Tiro.health, nexuzhealth

Data transport	FHIR server infrastructure	Receive Transaction Bundle, validate against StructureDefinitions, expose resources via standard FHIR search.	Amaron, nexuzhealth, Axian
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Test Scenarios

Test 1: Definition-based Extraction to FHIR Resources (core)

Objective: Transform a completed QuestionnaireResponse into discrete, searchable FHIR resources using the `.definition` element and the `$extract` operation.

1. **Setup:** Participants receive a FHIR Questionnaire where items are linked to profiled FHIR resources via `.definition` (e.g., items mapping to elements of an Observation profile, a Condition profile, or a Procedure profile).
2. **Input:** A completed QuestionnaireResponse (e.g., a home care nursing assessment or a registry submission).
3. **The Challenge:**
 - Call the `$extract` operation (using Tiro.health's public API or a participant's own implementation) to transform the QuestionnaireResponse into a FHIR Transaction Bundle.
 - Verify that fixed values, slicing, and patterns defined in the target profiles are correctly applied in the extracted resources.
4. **Success Criteria:** The generated Bundle is successfully POSTed to the eHealth Test Server, and the individual resources can be retrieved via standard FHIR search (e.g., `GET /Observation?patient=[ID]&code=[LOINC]`).

Test 2: Extraction to Logical Models (advanced)

Objective: Extract data from a QuestionnaireResponse into a custom data model defined as a FHIR logical model (StructureDefinition with `kind: logical`).

1. **Setup:** Participants receive (or define) a logical model representing a registry-specific data structure, and a Questionnaire with `.definition` elements pointing to paths in that logical model.
2. **Input:** A completed QuestionnaireResponse for the registry form.
3. **The Challenge:**
 - Call the `$extract` operation with an `Accept: application/json` header to trigger logical model extraction.
 - Validate that the returned Binary resource contains JSON that conforms to the logical model structure.
4. **Success Criteria:** The extracted JSON matches the logical model's element structure and can be consumed by a registry-specific system.

Test 3: Pre-population (bonus, if time permits)

Objective: Automatically populate a Questionnaire with existing patient data to reduce clinician data entry.

1. **Source Data:** Participants use the eHealth Test Server or their own FHIR server as the data source.
2. **The Challenge:** Use SDC expressions such as `initialExpression` or `launchContext` to define how existing data should be fetched and mapped to Questionnaire items.
3. **Success Criteria:** A QuestionnaireResponse is generated in "in-progress" status with fields correctly pre-filled from existing patient data.